

ROUEN VALLEE DE SEINE AP, France

WMO: 070370

Lat: 49.3831N

Lon: 1.1817E

Elev: 512

StdP: 14.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	26.0	17.1	13.3	26.5	20.5	15.7	27.9	29.7	49.4	26.7	49.0	7.6	40	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	84.6	67.8	80.7	66.4	77.1	64.7	69.5	81.2	67.5	78.0	65.8	74.7	9.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	95.9	74.1	63.7	90.0	71.6	62.2	85.3	69.7	33.8	81.1	32.2	78.7	30.8	75.0	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	20.7	18.5	18.6	92.0	4.0	4.9	15.7	95.5	13.3	98.4	11.0	101.2	8.1	104.7		
(5)			17.7	73.4	4.1	1.8	14.7	74.7	12.3	75.8	10.0	76.8	7.0	78.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	39.4	40.6	45.0	49.5	55.0	60.5	64.3	64.1	59.1	53.1	45.5	40.2	(6)
(7)		DBStd	10.74	6.93	6.51	6.12	6.05	5.92	5.71	5.56	5.52	5.08	5.72	5.94	6.79	(7)
(8)		HDD50	1368	329	266	175	80	17	1	0	0	2	34	157	307	(8)
(9)		HDD65	5107	792	683	619	465	313	159	81	83	190	370	585	767	(9)
(10)		CDD50	1891	2	3	21	65	174	316	445	437	273	130	21	4	(10)
(11)		CDD65	157	0	0	0	0	5	25	60	55	11	1	0	0	(11)
(12)		CDH74	1505	0	0	0	8	53	259	571	513	96	5	0	0	(12)
(13)		CDH80	415	0	0	0	0	3	65	163	168	16	0	0	0	(13)
(14)	Wind	WSAvg	9.3	10.7	10.6	10.2	9.3	8.9	8.3	8.1	7.7	8.2	9.3	9.6	10.2	(14)
(15)	Precipitation	PrecAvg	31.1	2.9	2.1	2.6	1.9	2.8	2.2	2.4	2.4	2.7	3.0	3.5	3.0	(15)
(16)		PrecMax	40.3	6.2	5.3	5.2	3.9	5.8	5.4	6.6	6.3	6.6	6.9	6.3	7.0	(16)
(17)		PrecMin	18.9	0.6	0.2	0.3	0.2	0.7	0.1	0.4	0.1	0.7	0.4	0.7	0.2	(17)
(18)		PrecStd	5.4	1.5	1.2	1.2	0.9	1.3	1.2	1.2	1.3	1.5	1.8	1.4	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	58.5	66.7	74.7	78.9	86.2	90.2	91.3	82.0	72.4	61.3	56.1	(19)
(20)			MCWB	52.6	50.5	53.8	59.7	65.3	69.2	69.5	69.8	66.0	62.8	57.7	53.5	(20)
(21)		2%	DB	53.2	53.8	61.6	69.1	75.0	80.9	84.4	84.4	76.4	67.0	58.0	54.1	(21)
(22)			MCWB	51.0	49.4	52.3	57.6	62.4	66.9	67.4	67.8	64.3	60.6	55.2	51.9	(22)
(23)		5%	DB	50.8	51.5	57.5	64.1	70.8	76.1	80.5	79.4	71.8	63.8	55.8	52.2	(23)
(24)			MCWB	48.7	48.3	50.7	54.4	60.4	64.5	66.2	66.0	61.9	58.7	53.4	50.2	(24)
(25)		10%	DB	48.8	49.6	54.2	60.0	66.5	72.0	76.5	75.3	68.4	61.3	54.0	50.2	(25)
(26)			MCWB	46.9	47.0	49.4	52.1	58.1	62.3	64.7	64.0	60.1	57.4	51.8	48.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	52.2	55.6	61.8	67.3	71.5	71.7	72.1	68.3	64.3	59.1	54.4	(27)
(28)			MCD B	54.6	55.3	62.1	72.5	77.0	83.0	86.4	85.6	78.2	70.0	60.5	55.4	(28)
(29)		2%	WB	51.4	50.4	53.6	58.2	63.8	68.0	69.2	69.2	65.4	61.6	55.8	52.3	(29)
(30)			MCD B	53.0	52.9	59.0	67.3	71.9	78.4	81.5	81.0	73.3	65.8	57.4	53.7	(30)
(31)		5%	WB	49.1	48.9	52.0	55.2	61.0	65.6	67.2	67.1	63.3	59.8	53.9	50.6	(31)
(32)			MCD B	50.6	51.1	56.5	62.2	68.7	74.6	77.6	77.5	69.8	63.0	55.6	52.1	(32)
(33)		10%	WB	47.1	47.1	50.1	52.9	58.8	63.3	65.3	65.1	61.6	58.1	52.1	48.5	(33)
(34)			MCD B	48.6	49.2	53.7	58.5	65.6	70.7	74.3	73.1	66.9	60.8	53.8	49.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.9	10.9	13.5	16.6	16.8	17.8	18.3	18.4	16.7	13.3	10.1	8.9	(35)
(36)			MCDBR	9.7	13.1	18.6	22.6	23.2	24.4	25.6	25.7	22.0	15.9	10.9	9.6	(36)
(37)			MCWBR	8.6	9.7	11.4	12.9	13.1	12.9	11.8	11.6	11.1	10.2	8.6	8.6	(37)
(38)		5% WB	MCD BR	9.3	11.5	15.6	20.0	20.6	22.2	23.2	22.9	18.6	14.2	10.2	9.4	(38)
(39)			MCWBR	8.7	9.3	10.4	12.0	12.4	12.8	11.7	11.4	10.4	10.0	8.9	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.338	0.372	0.393	0.400	0.400	0.397	0.390	0.368	0.355	0.329	0.314	(40)	
(41)		taud	2.426	2.391	2.290	2.245	2.261	2.293	2.302	2.345	2.388	2.428	2.439	2.398	(41)	
(42)		Ebn at Noon	231	255	263	270	272	273	272	268	262	245	226	215	(42)	
(43)		Edh at Noon	20	27	35	40	41	40	39	36	31	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	284	520	882	1363	1596	1786	1702	1453	1122	652	340	240	(44)	
(45)		RadStd	29	69	105	152	138	175	127	118	73	59	33	32	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (39 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page